

CTD QUIZ - 5 SOLUTION
(SET-A)

mag - 6 marks

Sol (1)÷ Given transfer function,

$$\begin{aligned} H(s) &= \frac{4 \times 10^6}{s^3 + 4100s^2 + (4 \times 10^5)s} \\ &= \frac{4 \times 10^6}{s(s^2 + 4100s + 4 \times 10^5)} \\ &= \frac{4 \times 10^6}{s(s+100)(s+4000)} \\ &= \frac{10}{s\left(\frac{s}{100} + 1\right)\left(\frac{s}{4000} + 1\right)} \quad \text{--- (1)} \end{aligned}$$

→ (1 marks)

Bode magnitude plot÷

$$|H(s)| = \left| \frac{10}{s} \times \frac{1}{\left(\frac{s}{100} + 1\right)} \times \frac{1}{\left(\frac{s}{4000} + 1\right)} \right|$$

$$\begin{aligned} |H(s)|_{dB} &= 20 \log_{10} \left| \frac{10}{s} \times \frac{1}{\left(\frac{s}{100} + 1\right)} \times \frac{1}{\left(\frac{s}{4000} + 1\right)} \right| \\ &= 20 \log_{10} \left| \frac{10}{s} \right| + 20 \log_{10} \left| \frac{1}{\left(\frac{s}{100} + 1\right)} \right| + 20 \log_{10} \left| \frac{1}{\left(\frac{s}{4000} + 1\right)} \right| \quad \text{--- (2)} \end{aligned}$$

the overall bode magnitude plot is equal to sum of bode magnitude plots of individual factors.

$$\begin{aligned} \text{(I)} \quad |H(s)|_{dB} &= 20 \log_{10} \left| \frac{10}{s} \right| = 20 \log_{10} \left(\frac{10}{\omega} \right) = (20 \log_{10} 10 - \\ &= (-20 \log_{10} \omega + 20) \text{ dB} \quad \text{--- (3)} \end{aligned}$$

Passes through the point $(1, 20 \log 10 \text{ dB}) = (1, 20 \text{ dB})$
 & having a slope of -20 dB/decade .

$$\begin{aligned}
 \text{(II)} \quad |H(s)_{II}|_{\text{dB}} &= 20 \log_{10} \left| \frac{1}{\left(\frac{s}{100} + 1\right)} \right| \text{ dB} \\
 &= 20 \log_{10} \left(\frac{1}{\sqrt{\left(\frac{\omega}{100}\right)^2 + 1}} \right) \text{ dB} \\
 &= -20 \log_{10} \sqrt{\left(\frac{\omega}{100}\right)^2 + 1} \text{ dB} \\
 &= -10 \log_{10} \left[\left(\frac{\omega}{100}\right)^2 + 1 \right] \text{ dB} \quad \text{--- (4)}
 \end{aligned}$$

Cut-off frequency, $\omega_c = 100 \text{ rad/sec}$

$$\begin{aligned}
 |H(s)_{II}|_{\text{dB}} &\cong -10 \log_{10} \left(\frac{\omega}{100}\right)^2 \text{ dB} \quad (\omega^2 \gg 1) \\
 &= (-20 \log_{10} \omega + 20 \log_{10} 100) \text{ dB} \\
 &= (-20 \log_{10} \omega + 40) \text{ dB} \quad \text{--- (5)}
 \end{aligned}$$

$$|H(s)_{II}|_{\text{dB}} \cong 0 \text{ dB} \quad \text{--- (6)} \quad (\omega^2 \ll 1)$$

Passes through 0 dB -axis upto the cut-off freq ($\omega_c = 100 \text{ rad/sec}$) & having a slope of -20 dB/decade after the cut-off freq.

$$\begin{aligned}
 \text{(III)} \quad |H(s)_{III}|_{\text{dB}} &= 20 \log_{10} \left| \frac{1}{\left(\frac{s}{4000} + 1\right)} \right| \text{ dB} \\
 &= 20 \log_{10} \left(\frac{1}{\sqrt{\left(\frac{\omega}{4000}\right)^2 + 1}} \right) \text{ dB}
 \end{aligned}$$

$$= -20 \log_{10} \sqrt{\left(\frac{\omega}{4000}\right)^2 + 1} \text{ dB}$$

$$= -10 \log_{10} \left[\left(\frac{\omega}{4000}\right)^2 + 1 \right] \text{ dB} \quad \text{--- (7)}$$

corner freq, $\omega_c = 4000 \text{ rad/sec}$

$$|H(s)|_{\text{dB}} = 0 \text{ dB} \quad \text{--- (8)} \quad (\omega^2 T^2 \ll 1)$$

$$|H(s)|_{\text{dB}} = -10 \log_{10} \left(\frac{\omega}{4000}\right)^2 \text{ dB} \quad (\omega^2 T^2 \gg 1)$$

$$= (-20 \log_{10} \omega + 20 \log_{10} 4000) \text{ dB}$$

$$= (-20 \log_{10} \omega + 72.04) \text{ dB} \quad \text{--- (9)}$$

Passes through 0-dB axis upto the corner freq ($\omega_c = 4000 \text{ rad/sec}$) & having a slope of -20 dB/decade after the ω_c .

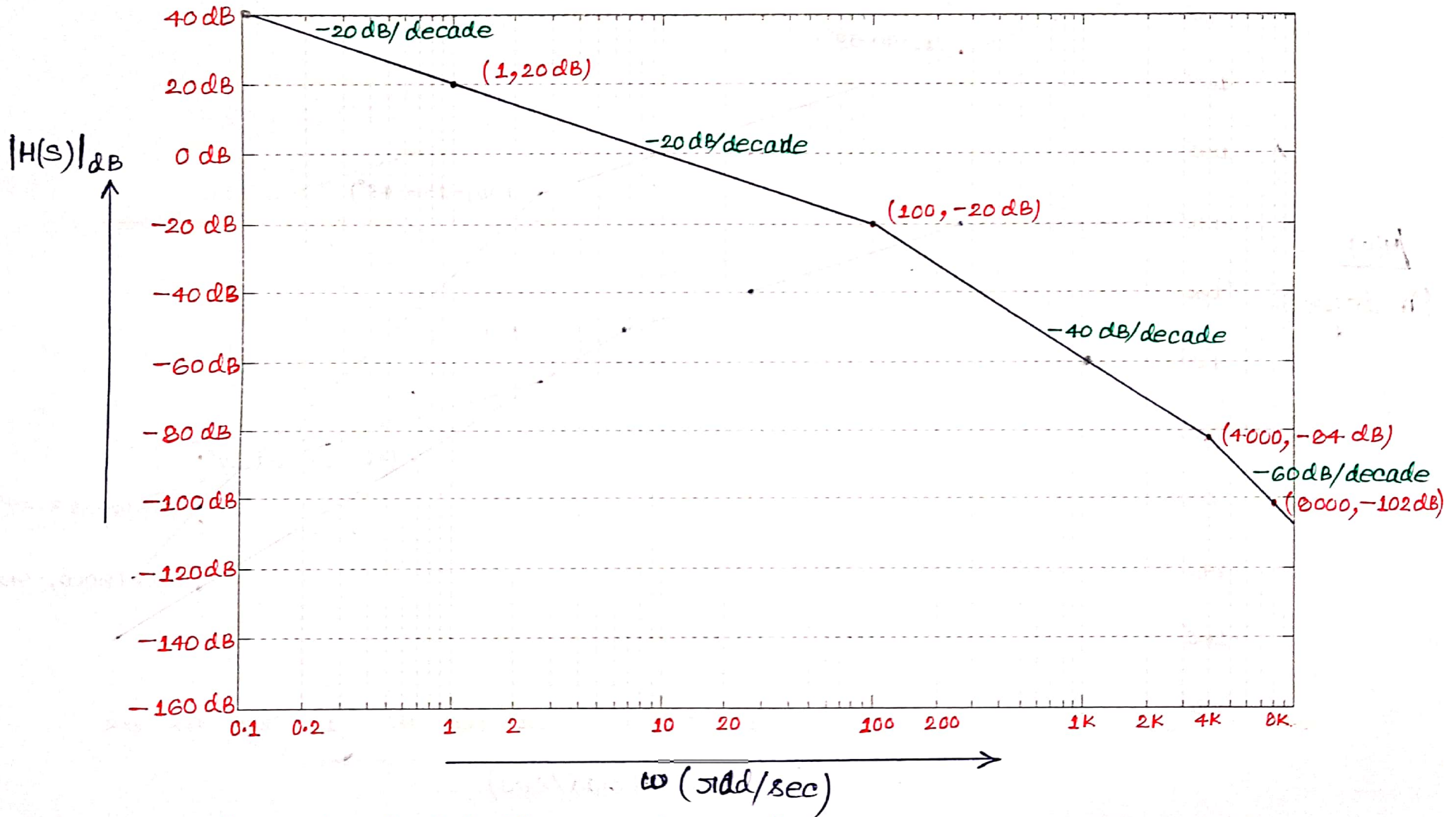
Bode phase plot ÷

$$\angle H(s) = -90^\circ - \tan^{-1}\left(\frac{\omega}{100}\right) - \tan^{-1}\left(\frac{\omega}{4000}\right)$$

Semi-log Paper for Bode Plots

Bode Magnitude Plot (Set-A) :

Indivi. graphs $\Rightarrow$ 3 Marks
Final " $\Rightarrow$ 1 Mark

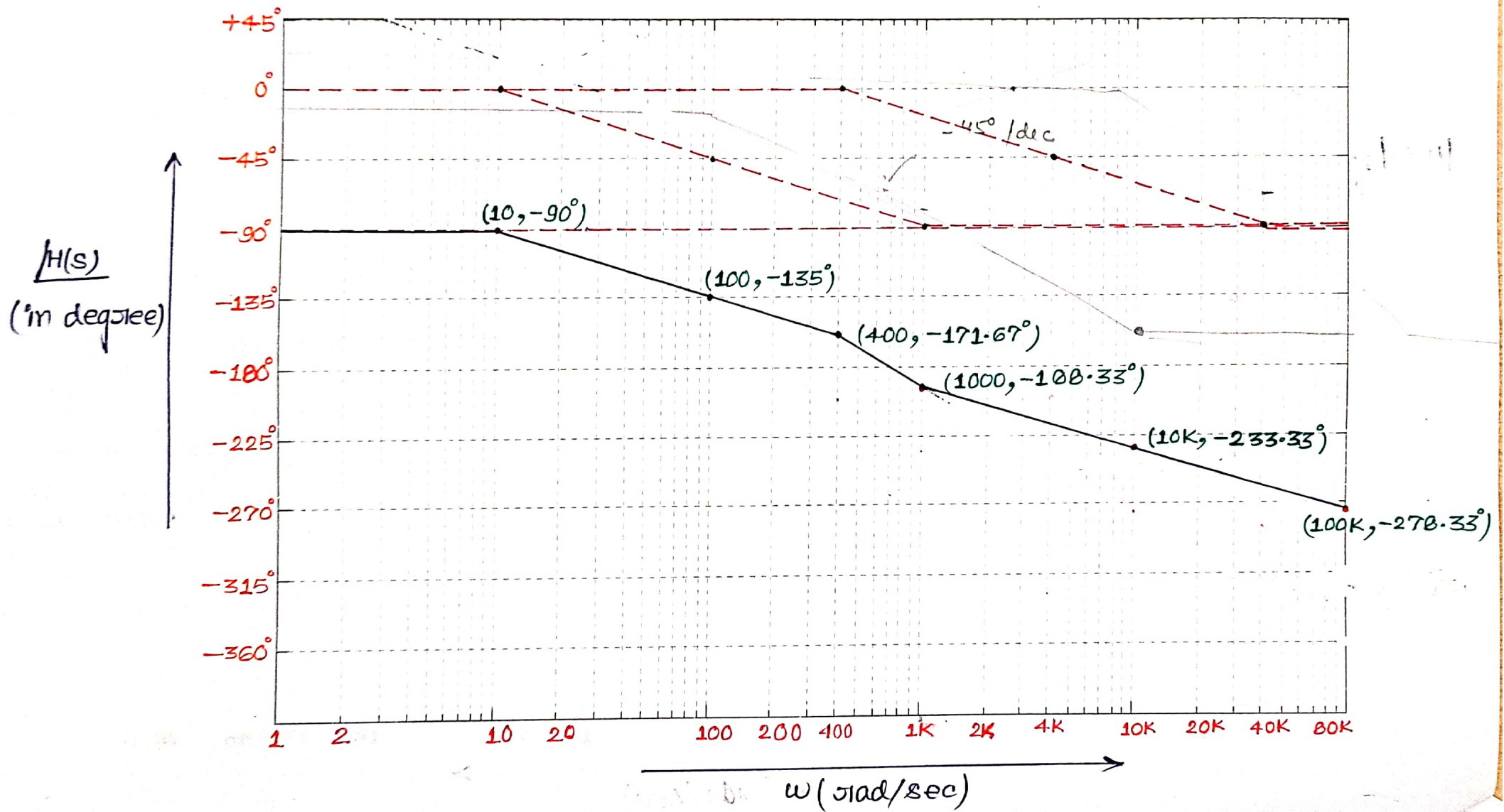


Correct axis labeling $\Rightarrow$ (1 Mark)

Semi-log Paper for Bode Plots

Bode Phase Plot (Set-A)

Indivi. graph $\Rightarrow$ 3 Marks
Final graph $\Rightarrow$ 1 Mark



CTD QUIZ-5 SOLUTION
(SET-B)

Sol(1)÷ Given transfer function,

$$\begin{aligned} H(s) &= \frac{3000s}{s^2 + 3100s + 3 \times 10^5} \\ &= \frac{3000s}{(s+100)(s+3000)} \\ &= \frac{0.01s}{\left(\frac{s}{100} + 1\right)\left(\frac{s}{3000} + 1\right)} \quad \text{--- (1)} \end{aligned}$$

→ (1 Mark)

Bode Magnitude plot÷

$$\begin{aligned} |H(s)|_{dB} &= 20 \log_{10} \left| 0.01s \times \frac{1}{\left(\frac{s}{100} + 1\right)} \times \frac{1}{\left(\frac{s}{3000} + 1\right)} \right| \\ &= 20 \log_{10} |0.01s| + 20 \log_{10} \left| \frac{1}{\left(\frac{s}{100} + 1\right)} \right| + 20 \log_{10} \left| \frac{1}{\left(\frac{s}{3000} + 1\right)} \right| \end{aligned}$$

--- (2)

$$(I) |H(s)_I|_{dB} = 20 \log_{10} |0.01s| = 20 \log_{10} (0.01\omega) \text{ dB} \quad \text{--- (3)}$$

Passes through the point $(1, -40 \text{ dB})$ & having a slope of $+20 \text{ dB/decade}$.

$$(II) |H(s)_II|_{dB} = 20 \log_{10} \left| \frac{1}{\left(\frac{s}{100} + 1\right)} \right| = -10 \log_{10} \left[\left(\frac{\omega}{100}\right)^2 + 1 \right] \text{ dB}$$

--- (4)

Cut-off freq, $\omega_c = 100 \text{ rad/sec}$

$$|H(s)_II|_{dB} = 0 \text{ dB} \quad \text{--- (5)} \quad (\omega^2 T^2 \ll 1)$$

$$|H(s)_II|_{dB} = (-20 \log_{10} \omega + 40) \text{ dB} \quad \text{--- (6)} \quad (\omega^2 T^2 \gg 1)$$

Passes through 0 dB-axis upto the corner freq ($\omega_c = 100$ rad/sec) & having a slope of -20 dB/decade after that.

$$\begin{aligned} \text{(III)} \quad |H(s)_{\text{III}}|_{\text{dB}} &= 20 \log \left| \frac{1}{\left(\frac{s}{3000} + 1\right)} \right| \text{ dB} \\ &= -10 \log_{10} \left[\left(\frac{\omega}{3000}\right)^2 + 1 \right] \text{ dB} \quad \text{--- (7)} \end{aligned}$$

corner freq, $\omega_c = 3000$ rad/sec

$$|H(s)_{\text{III}}|_{\text{dB}} = 0 \text{ dB} \quad \text{--- (8)} \quad (\omega^2 T^2 \ll 1)$$

$$|H(s)_{\text{III}}|_{\text{dB}} = (-20 \log_{10} \omega + 69.54) \text{ dB} \quad \text{--- (9)} \quad (\omega^2 T^2 \gg 1)$$

Passes through 0 dB-axis upto the corner frequency ($\omega_c = 3000$ rad/sec) & having a slope of -20 dB/decade after ω_c .

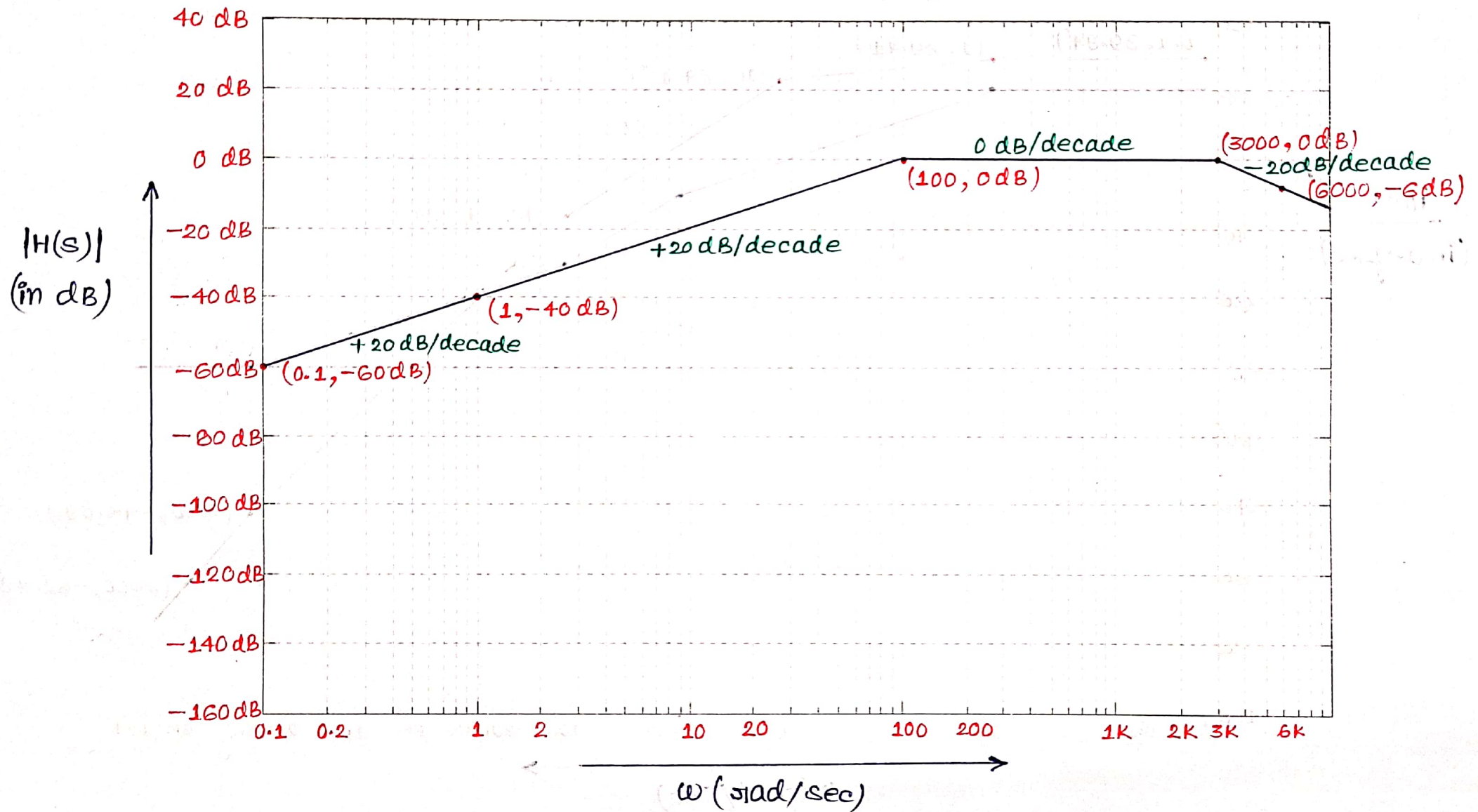
Bode phase plot $\frac{\circ}{\circ}$

$$\angle H(s) = 90^\circ - \tan^{-1} \left(\frac{\omega}{100} \right) - \tan^{-1} \left(\frac{\omega}{3000} \right)$$

Semi-log Paper for Bode Plots

Bode Magnitude Plot (Set-B)

Indi graph $\Rightarrow$ 3 Marks
Final graph $\Rightarrow$ 1 Mark.



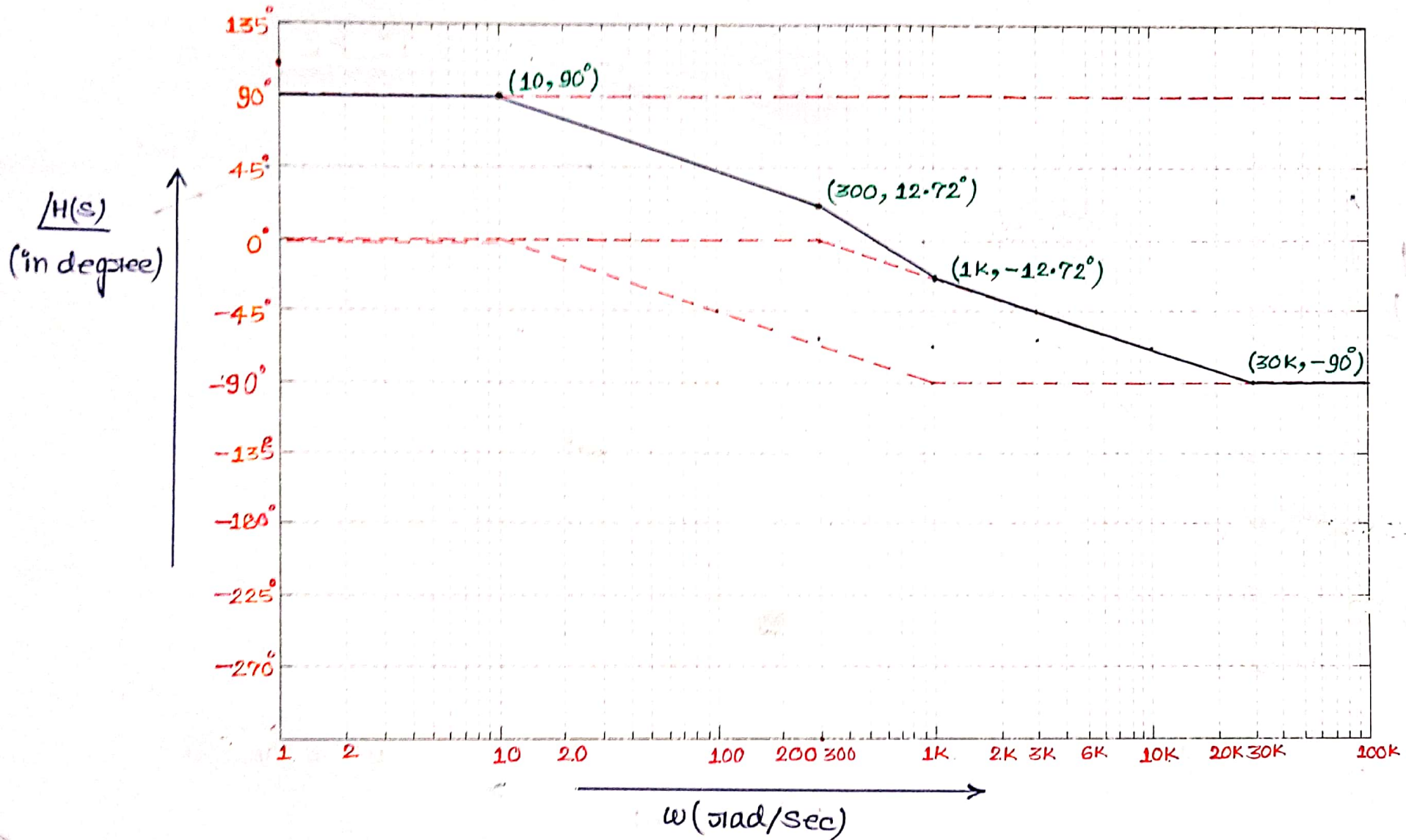
90°

Semi-log Paper for Bode Plots

Bode Phase Plot (Sec-B)

Indi-graph $\Rightarrow$ 3 Marks

Final " $\Rightarrow$ 1 Mark



Correct axis labels $\Rightarrow$ (1 Mark)

